Advancements in X-Ray Diffraction and X-Ray Free-Electron Lasers

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Due to their nanometer-scale wavelengths, X-rays are well-suited for probing atomic-scale structures. This paper discusses the leaps in X-ray source brilliance made possible by particle accelerator technology, leading to synchrotron lasers and eventually X-ray free-electron lasers. Further, the effect such X-ray sources have had on X-ray diffraction (XRD) experiments is studied, providing a sufficient background for the theoretical underpinnings of crystallography, which allows for the exploration of how techniques such as serial femtosecond crystallography (SFX) and single-particle imaging (SPI) allows scientists to push the spatial resolution of crystal structure determination down to the nanometer scale. The temporal characteristics of these powerful pulses are similarly described to understand how these facilities push the envelope of ultrafast science through pump-probe experiments allowing for real-time imaging of chemical reactions and phase transitions which is of significant interest to various fields, including chemistry, biology, crystallography, and much more. Of course, no technique is free of technical challenges and limitations, and these too will be explored.

Contents

I. INTRODUCTION

X-ray diffraction (XRD) has been instrumental in elucidating atomic structures since its experimental validation in 1912 by Friedrich, Knipping, and von Laue [1]. X-ray tubes have been the conventional laboratory source of such radiation, due to their simple assembly and ease of maintenance. Such tubes have a very low overall efficiency, however, with most of the energy supplied to the tube being lost due to heat, and therefore requiring continuous cooling [2]. These sources also have a relatively low intensity, and emit incoherent light. **F**ree-**E**lectron **L**asers (FELs), introduced by John Madey in the early 70's, operated at infrared, visible and ultraviolet wavelengths [3]. In 2005, the **D**eutsches **E**lektronen-**SY**nchrotron's (DESY) **F**ree-electron **LAS**er in **H**amburg (FLASH), Germany, became the first facility to produce X-rays using this apparatus, with SLAC's **L**inac **C**oherent **L**ight **S**ource (LCLS) following suit with higher energy X-rays in 2009. Various other facilities have been constructed around the world since. Such advancements in X-ray sources have led to a revolution in X-ray diffraction, allowing for the determination of protein structures using microcrystals, the emerging field of serial femtosecond crystallography (SFX), and the study of ultrafast phenomena at the atomic scale, such as bond formation and breaking during chemical reactions.

II. THEORETICAL FRAMEWORK

A. Bragg's Law

Shortly after Röntgen's discovery of X-rays, Max von Laue suggested that X-rays could be diffracted by crystals, which Friedrich and Knipping experimentally confirmed in 1912. The underlying principle was formalized by W. H. Bragg and W. L. Bragg through:

$$2d\sin\theta = n\lambda, \tag{1}$$

where d is the regular spacing between the atoms, θ the angle of incidence, n an integer, and λ the X-ray wavelength [4]. As an X-ray beam incident upon a crystal interacts with the electrons of the lattice's atoms, it is scattered spherically in all directions. The scattered X-rays from adjacent atoms interfere constructively and destructively, producing a diffracted beam. The condition for constructive interference is given by Eq. 1. That is, when the path difference between two adjacent rays ($2d\sin\theta$) is an integer multiple of the wavelength, the waves will superimpose in phase, leading to peaks in the diffracted intensity. This phenomenon is illustrated in Fig. 2.

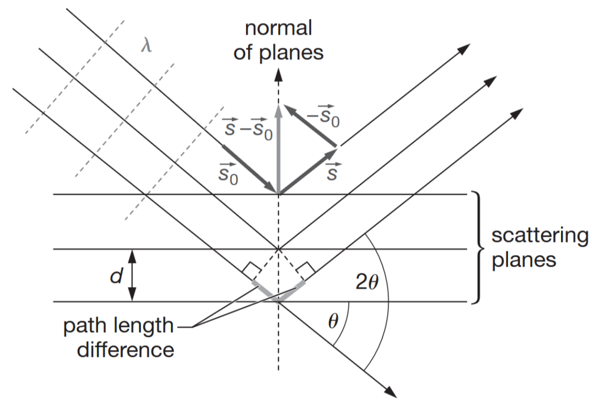

FIG. 1. Geometry of a beam with angle of incidence θ approaching from the left into a symmetrical arrangement of atoms forming the lattice of a crystal [5].

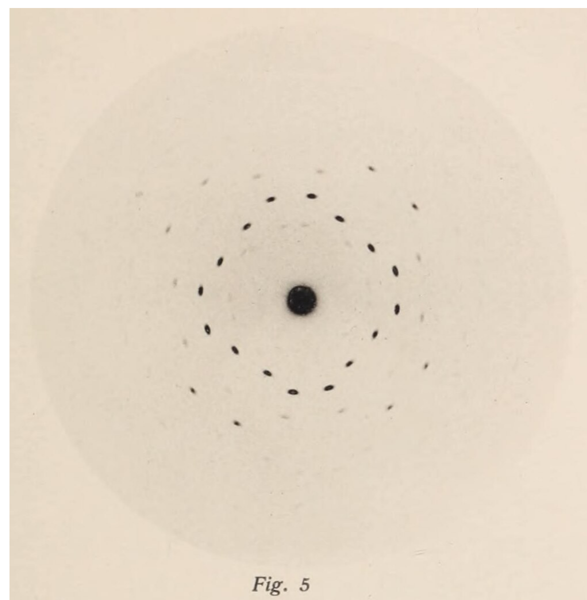

FIG. 2. X-ray diffraction pattern of a copper sulfate crystal from Friedrich and Knipping's 1912 paper [1].

B. The Structure of Crystals

The theory of crystals and how they are structured is essential to understanding X-ray diffraction. Crystals are regular arrangements of atoms in three dimensions, exhibiting translational symmetry. The discrete points at which the atoms of a crystal are located can be described mathematically by

$$\mathbf{R} = n_1\mathbf{a}_1 + n_2\mathbf{a}_2 + n_3\mathbf{a}_3, \tag{2}$$

where $\mathbf{a}_1$, $\mathbf{a}_2$, and $\mathbf{a}_3$ are the primitive vectors of the lattice and n_1, n_2, and n_3 are integers. This infinite set of points defined by Eq. 2 forms a Bravais lattice, and the parallelepiped comprised of these primitive vectors is called the unit cell. The constraint that crystals exhibit translational symmetry and repeat periodically in space leads to the classification of Bravais lattices into 14 distinct types [6]. These lattices can be further categorized into seven crystal systems based on the symmetry of the unit cell. The seven crystal systems are cubic, tetragonal, orthorhombic, hexagonal, rhombohedral, monoclinic, triclinic, and can be seen in Fig. 3. The concept of a reciprocal lattice, which is especially useful in X-ray diffraction, is defined as the set of all wave vectors $\mathbf{k}$ that yield plane waves with the periodicity of the Bravais lattice [7]. The reciprocal lattice itself is spanned by three primitive vectors $\mathbf{b}_1$, $\mathbf{b}_2$, and $\mathbf{b}_3$ which can be generated from the lattice's primitive vectors via:

$$\mathbf{b_i} = 2\pi \frac{\mathbf{a}_j \times \mathbf{a}_k}{\mathbf{a_i} \cdot (\mathbf{a}_j \times \mathbf{a}_k)}, \tag{3}$$

For any cyclic permutation of $i, j, k = 1, 2, 3$. The triple scalar product in the denominator is physically interpreted as the volume of the unit cell. The factor of 2π ensures that the reciprocal lattice vectors are properly scaled so that plane waves match the periodicity of the crystal. The reciprocal lattice vector is then defined as:

$$\mathbf{G} = h\mathbf{b}_1 + k\mathbf{b}_2 + l\mathbf{b}_3, \tag{4}$$

where h, k, and l are integers, commonly known as the Miller indices. $\mathbf{G}$ is perpendicular to the family of planes indexed by (hkl), these being the planes that diffract the incident X-rays. The inter-planar spacing d for a cubic crystal, for example, is given by:

$$d = \frac{a}{\sqrt{h^2 + k^2 + l^2}}, \tag{5}$$

where a is the lattice parameter, which represents the length of the unit cell's sides. For a cubic crystal, we have $a = |\mathbf{a}_1| = |\mathbf{a}_2| = |\mathbf{a}_3|$. For other crystal systems (e.g., tetragonal, orthorhombic), the interplanar spacing depends on a more general relation between the Miller indices, the lattice parameters, and the metric tensor of the crystal system [8].

Crystal System	**Length**	**Angle**
Triclinic	$a \neq b \neq c$	$\alpha \neq \beta \neq \gamma \neq 90°$
Monoclinic	$a \neq b \neq c$	$\alpha = \gamma = 90°$; $\beta \neq 90°$
Orthorhombic	$a \neq b \neq c$	$\alpha = \beta = \gamma = 90°$
Tetragonal	$a = b \neq c$	$\alpha = \beta = \gamma = 90°$
Trigonal (hexagonal)	$a = b \neq c$	$\alpha = \beta = 90°$; $\gamma = 120°$
Trigonal (rhombohedral)	$a = b = c$	$\alpha = \beta = \gamma \neq 90°$
Hexagonal	$a = b \neq c$	$\alpha = \beta = 90°$; $\gamma = 120°$
Cubic	$a = b = c$	$\alpha = \beta = \gamma = 90°$

TABLE I. The seven crystal systems and the geometric properties of their unit cells [10].

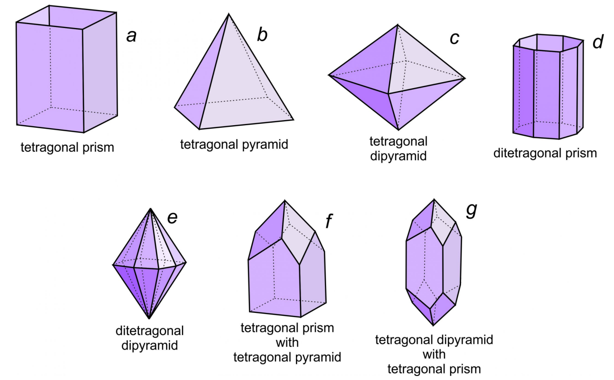

FIG. 3. The seven 3D crystal systems and their corresponding unit cells [9].

III. EXPERIMENTAL APPARATUS

Modern XRD experiments use high-brightness sources such as synchrotrons and XFELs to generate X-rays. X-ray sources are characterised by their brightness, which is the ratio of the intensity (photons per second) of the beam to its spatial coherence, which is proportional to the product of the size (m^2), bandwidth and angular divergence of the beam. The difference in brightness between XFELs and synchrotrons is drastic, with XFELs exhibiting higher brightness by several orders of magnitude due to their high peak power and short pulse duration [11]. A comparison of the average brightness of XFEL and synchrotron facilities is shown in Fig. 4.

A. Electron Guns

In order to accelerate electrons to relativistic speeds, we first need a way to generate them. The electron gun is the start of the story for this X-ray source. These electron sources typically use thermionic emission or field emission methods. In thermionic emission, a heated filament releases electrons, while in field emission, a strong electric field extracts electrons from a sharp metal tip. Once emitted, the electrons are accelerated by a high-voltage anode, typically to energies ranging from 5 to 20 MeV, depending on the XFEL facility.

B. Linear Accelerators

XFELs and synchrotrons operate in a similar manner, accelerating electrons to relativistic speeds and inducing synchrotron radiation via the Lorentz force. Synchrotrons use bending magnets to curve the electron beam's trajectory, while XFELs use undulators installed at the end of a linear accelerator (LINAC) to generate coherent X-rays, though some synchrotrons have also implemented undulators as insertion devices. Both sources

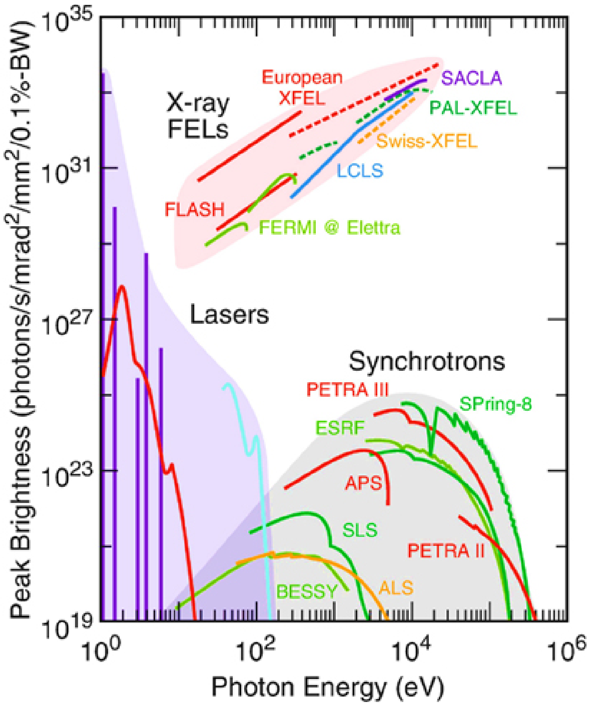

FIG. 4. Comparison of peak brightness as a function of photon energy between conventional laser sources and higher synchrotons, XFELs and harmonic generation sources, on a log-log scale. [12].

require that electron bunches be accelerated under vacuum, where the air pressure is of the order of 10^{-9} Torr to prevent beam scattering and energy loss. An initial, naive concept for accelerating charges to such speeds would be to upscale the parallel-plate capacitor, building a large potential difference across the plates and thus accelerating electrons. While such devices do exist, called Van De Graaff accelerators, they suffer from being too unstable when scaled up to the level required to meet current energy thresholds, discharging (dangerously!) with surrounding air. Thus, modern linear accelerators make use of more sophisticated technology in the form of radio-frequency (RF) cavities. These work via radio frequency electromagnetic pulses (often in the GHz range) sent from a generator via a waveguide into spheroidal resonant cells which together form a cavity. These waves resonate in the various cells, forming standing waves. The system is engineered such that any two neighbouring cells are 180° out of phase. Thus, through precise timing, an electron bunch can be sent through the first cell during its forward-thrusting phase, and by the time it reaches the next cell, the standing wave pattern inverts and this cell, too, is forward thrusting. This repeats ad nauseum until the electrons are at the desired velocity. Of course, adjacent cells must be of increasing length towards the start of the LINAC, such that as electrons increase in velocity, traversing longer distances in the same amount of time, the arrival of the electrons to the new cells are still synchronized with the standing wave pattern. Towards the end of the LINAC, where electrons are in the relativistic regime, speed increases become negligible and this ceases to be the case.

In state-of-the art XFEL facilities, such as EuXFEL and LCLS-II, superconducting radio-frequency (SRF) cavities are in use (often Niobium-based), kept at approximately two degrees kelvin. With this, dissipative losses due to heat induced by surface currents as a result of the electromagnetic waves in the cavity are astonishingly low. Thish drastically increases the resonator's quality factor, defined as the ratio of stored energy to the energy lost per cycle. It is often said that had Galileo, being one of the first to study mechanical resonance in a pendulum in the 17$^{\text{th}}$ century, worked with such a resonator operating at 1 Hz, this very pendulum would still be swinging with about half its original amplitude [13]. Such a high quality resonator enables continuous-wave operation, in which the RF field in the cavity is maintained at a steady amplitude for long periods of time. This allows for multiple bunches to be accelerated with little-to-no down-time. Thus, the repetition rate of the aforementioned facilities is in the millions of electron bunches per second. Many other facilities, lagging behind on the adoption of SRF technology, are undergoing upgrades in accelerator design to meet such repetition rate demands, thus enabling ultrafast science.

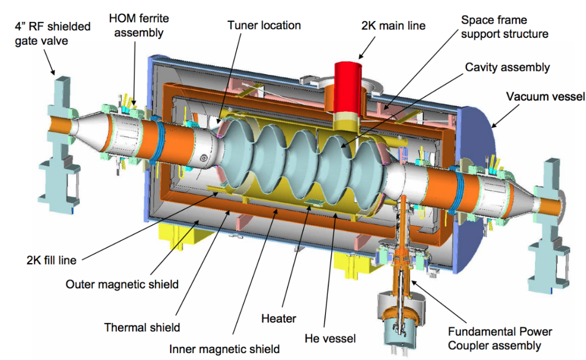

FIG. 5. Superconducting radio-frequency cavity schematic with supporting He cooling system and vacuum [13].

C. Undulators

Undulators, as can be seen in fig 6, are arrays of magnets of periodically alternating polarities. As an electron enters this array at velocity $\mathbf{u} \approx c$, it undergoes synchrotron radiation due to the acceleration caused by the sinusoidal magnetic field. In the electron's centre of mass frame, where the undulator approaches at speed $-\mathbf{u}$, the undulator transverse $\mathbf{B}$ field becomes the combination of a transverse $\mathbf{B}$ field and a transverse $\mathbf{E}$ field via the

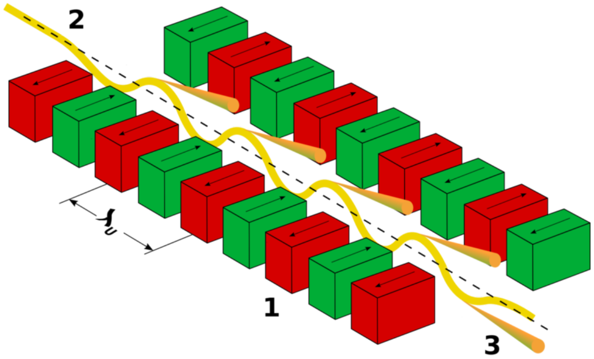

FIG. 6. Undulator schematic. Pictured are (1) two rows of adjacent magnets of opposing polarities with spacing $\lambda_u/2$, (2) incident beam of electrons and (3) emitted radiation.

Lorentz boosts:

$$\mathbf{E}'_{\parallel} = \mathbf{E}_{\parallel}, \quad \mathbf{B}'_{\parallel} = \mathbf{B}_{\parallel} \tag{6}$$

$$\mathbf{E}'_{\perp} = \gamma(\mathbf{E}_{\perp} - u\mathbf{B}_{\perp}) \tag{7}$$

$$\mathbf{B}'_{\perp} = \gamma\left(\mathbf{B}_{\perp} + \frac{u}{c^2}\mathbf{E}_{\perp}\right) \tag{8}$$

where $\gamma = \frac{1}{\sqrt{1-u^2/c^2}}$. Since the undulator field in the lab frame is purely magnetic ($\mathbf{E} = 0$), the transverse field in the electron's rest frame simplifies to $\mathbf{E}'_{\perp} = -\gamma u \mathbf{B}_{\perp}$. In essence, the electron is accelerated side-to-side in its rest-frame, undergoing dipole-like oscillations, which manifests itself as photons of wavelength equal to the undulator period λ_u, corrected for length contractions ($\lambda' = \lambda_u/\gamma$). Then, the emitted radiation is Doppler-shifted to a wavelength λ in the lab frame according to the standard formula

$$\lambda = \lambda' \sqrt{\frac{1-\beta}{1+\beta}} \tag{9}$$

$$= \lambda' \sqrt{\frac{1-\beta}{1+\beta}} \cdot \sqrt{\frac{1+\beta}{1+\beta}} \tag{10}$$

$$= \lambda' \frac{\sqrt{1-\beta^2}}{1+\beta} = \frac{\lambda'}{\gamma(1+\beta)} \tag{11}$$

$$\tag{12}$$

where $\beta = \frac{u}{c}$, which we can approximate as $\beta \approx 1$ for the relativistic electrons in the undulator. This, alongside the fact that $\lambda' = \lambda_u/\gamma$, leads to $\lambda = \lambda_u/(2\gamma^2)$, which is the wavelength of the emitted radiation in the lab frame. This fails to account for the fact that the electron's motion within the undulator is not purely longitudinal. The alternating magnetic field of the undulator imparts a transverse oscillatory velocity to the electron, such that in addition to moving forward, it wiggles side-to-side in a sinusoidal fashion. Without going into the details of the derivation, this leads to a correction of the form:

$$\lambda = \frac{\lambda_u}{2\gamma^2}\left(1 + \frac{K^2}{2} + \theta^2\gamma^2\right) \tag{13}$$

where,

$$K = \frac{eB_0\lambda_u}{2\pi m_e c}. \tag{14}$$

is known as the deflection parameter and is equal to the maximum angular excursion of the beam in units of $1/\gamma$, and θ is the observation angle, which is the angle between the undulator axis and the observation axis, describing how the wavelength is prone to angular doppler shifts. Any device having $K \lesssim 1$ is known as an undulator, while one with $K \gg 1$ is called a wiggler [14]. An undulator operates under relatively weak fields, causing the electron beam to undergo small transverse oscillations. These gentle wiggles (with undulator parameter $K \lesssim 1$) lead to coherent interference of emitted radiation, producing a narrow, highly collimated, and nearly monochromatic spectrum, ideal for XFELs. A wiggler, on the other hand, employs stronger magnetic fields, resulting in larger electron deflections ($K \gg 1$). The radiation from each oscillation cycle is emitted into different directions and does not interfere coherently, yielding a broader, more intense spectrum more akin to synchrotron radiation. Thus, while both devices use periodic magnetic fields, the undulator favors coherence and narrow spectral width, whereas the wiggler prioritizes brightness over spectral purity.

Equation 13 is handy as it entails that the emitted wavelength of an XFEL may be tuned by adjusting the magnetic field strength. It also reveals how much energy is required to accelerate electrons to sufficient speeds such that X-rays are emitted.

In reality, XFELs emit light with a bandwidth $\Delta\lambda$ centered around the wavelength given by equation 13. This is due the fact that each of the N_u oscillations of the electron's sinusoidal motion contributes a wave packet of finite duration. The total physical length of the resulting wave train is thus $L_t = N_u\lambda_u$, which leads to a wave train duration of $T_t \approx L_t/c = N_u\lambda_u/c$. Since the Fourier transform of a finite-length wave scales as $1/T_t$, the spectral width becomes

$$\frac{\Delta\lambda}{\lambda} = \frac{1}{N_u}, \tag{15}$$

or $1/mN_u$ for the m^{th} harmonic [14]. XFELs emit polarized X-rays due to the transverse acceleration of relativistic electrons. In planar undulators, where electrons oscillate in a single plane, the emitted radiation is linearly polarized. Observing off-axis may yield elliptical polarization, but the intensity is low due to strong relativistic beaming. For efficient generation of circular or elliptical polarization, dedicated devices such as elliptical undulators are employed [15].

D. Microbunching and Self-Amplified Spontaneous Emission (SASE)

In reality, it is not a single electron that is accelerated into the undulator, but groups of them called bunches. A bunch of electrons will emit X-rays as it passes through the undulator, but the radiation will be incoherent due to the random phase of the electrons.

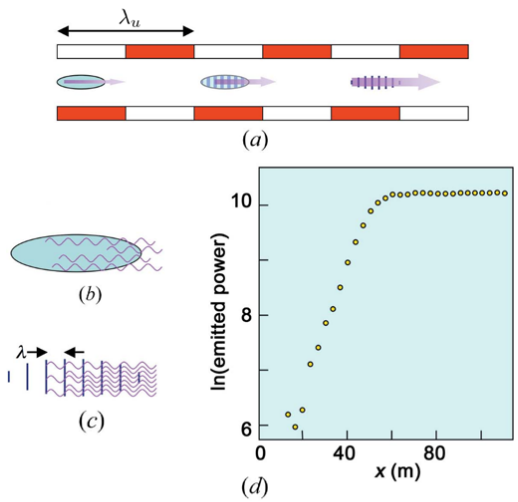

FIG. 7. Mechanism of microbunching in an XFEL. (a) shows the progressive formation of microbunches as the electrons pass through the undulator. (b) shows the initial waves emitted by the electrons, and (c) shows how the spacing between the microbunches is determined by the undulator period. Finally, (d) shows the how the intensity of the emitted X-rays scales as a function of distance in the undulator, until the saturation point is reached, corresponding to where the electrons are maximally bunched [16].

As relativistic electron bunches traverse the XFEL's undulator, their transverse motion causes them to emit synchrotron radiation. This emitted wave co-propagates with the electrons and interacts with them, giving rise to ponderomotive forces. These forces emerge from the coupling of the wave's electric field $\vec{E}_w$ and the electrons' transverse velocity $\vec{v}_T$ induced by the undulator magnetic field. Though both act transversely, their interplay results in a longitudinal average force:

$$\vec{f}_p \propto \vec{v}_T \times \vec{B}_w \tag{16}$$

Crucially, depending on the phase between $\vec{E}_w$ and $\vec{v}_T$, some electrons gain forward momentum, others lose it. This separates the bunch into density peaks and troughs (microbunches), spaced by the radiation wavelength λ. One might fear that the oscillation of $\vec{v}_T$ would cancel the effect after half an undulator period, as this velocity would reverse, and the electrons would be pushed out of their microbunches. This is not the case, as relativistic electrons travel just slightly slower than light so they slip the undulator period. This half-wavelength slip ensures that when $\vec{v}_T$ reverses, so do the wave fields, maintaining the same relative orientation, and thus preserving the bunching force. Through this mechanism, an initially uniform bunch evolves into a train of coherent emitters, and their in-phase radiation is exponentially amplified as it travels down the undulator [17], [18]. This correlated emission of X-rays leads to a significant increase in the intensity of the emitted radiation, as the waves emitted by the electrons constructively interfere. This is illustrated in Fig. 6. The intensity of the radiation scales exponentially as a function of distance in the undulator, as described by the following functional form:

$$I(z) = I_0 \exp\left(\frac{z}{l_g}\right), \tag{17}$$

where I_0 is the initial intensity of the radiation, z is the distance in the undulator, and l_g is the gain length. When building an undulator system, this result informs the optimal length to achieve the maximal theoretical intensity, with lengths exceeding those which achieve saturation proving to be unnecessary and cost-ineffective.

The advantages SASE has towards high peak brilliances can be understood by considering the total electric field produced by the bunch when each electron has a random phase ϕ_i. The total electric field produced is simply the real component of the superposition of each individual wave,

$$E_{tot} = \Re\left\{E_0 e^{i\omega t}\sum_{i=0}^{N} e^{i\phi_i}\right\}. \tag{18}$$

The root-mean-square amplitude, describing the average power, is thus dependent upon the sum of squares of unit phasors

$$Z^2 = \left\langle\left|\sum_{i=0}^{N} e^{i\phi_i}\right|^2\right\rangle = \left\langle\sum_{i=0}^{N}\sum_{j=0}^{N} e^{i(\phi_i-\phi_j)}\right\rangle. \tag{19}$$

We can easily compute this sum by considering the $i \neq j$ off-diagonal instances separately. When this holds, we are effectively summing over unit phasors of random orientations in the complex plane, and thus the ensemble average vanishes. Conversely, when $i = j$, the phases subtract to zero in the exponent and so the sum overall evaluates to N. Thus, we find that the average power of such a beam scales linearly with the number of electrons in the bunch. Now, consider an electron bunch all radiating with identical phase ϕ. The total electric field is trivially a superposition of N identical waves, which immediately gives $E_{tot} = \Re\{NE_0 e^{i(\omega t+\phi)}\}$. Squaring this quantity, to obtain the power, we see that this now scales by the *square* of the number of electrons. Noting that this quantity is often in the hundreds of *millions* to *billions*, we can appreciate the striking influence coherence has in brilliance contrasts between synchrotrons versus other X-ray sources.

E. Seeding

Free-electron lasers operating under the self-amplified spontaneous emission (SASE) regime are capable of producing extremely bright and ultrashort X-ray pulses. However, the stochastic nature of the SASE startup process results in significant shot-to-shot fluctuations, resulting in poor longitudinal coherence. These variations stem from the fact that amplification begins from random electron density noise, leading to poor longitudinal coherence and an unpredictable spectral profile. To address this limitation, several seeding techniques have been developed, aiming to impose an initial coherent signal onto the electron beam before significant amplification occurs. This coherent seed acts to guide the electron microbunching and radiation growth more deterministically, thus reducing fluctuations and improving the overall spectral quality of the emitted X-rays.

One widely used technique in the vacuum ultraviolet (VUV) and soft X-ray regimes is high-gain harmonic generation (HGHG). In HGHG, an external seed laser, such as a Ti:sapphire system frequency-converted into the UV or VUV, is introduced into a modulator undulator where it interacts with the relativistic electron bunch. This interaction induces an energy modulation in the electron beam at the seed frequency. A dispersive section then converts this energy modulation into a density modulation—creating periodic microbunches at the fundamental or harmonic wavelengths. These microbunched electrons then pass through a radiator undulator tuned to a harmonic of the seed wavelength, where coherent emission is amplified. Because the radiation is coherently initiated, HGHG offers excellent spectral brightness and stability, with narrower bandwidths than typical SASE sources.

However, HGHG becomes increasingly challenging at shorter wavelengths, especially in the hard X-ray regime, where suitable seed lasers are not readily available. This is where self-seeding techniques become more effective. Rather than injecting an external laser, self-seeding uses a monochromatized portion of the XFEL's own spontaneous SASE radiation as the seed. The electron beam first passes through an initial undulator section, where SASE is allowed to develop weakly. The resulting radiation is then filtered through a monochromator, typically a crystal for hard X-rays or a grating for soft X-rays, which selects a narrow bandwidth component. This filtered radiation is then recombined with the same electron bunch after a delay line, usually implemented with a magnetic chicane. The electrons and seed are resynchronized in the second undulator section, where amplification proceeds with improved temporal coherence [19]. While self-seeding does not reach the coherence levels achievable with external laser seeding, it has the advantage of being implementable at X-ray wavelengths far beyond the reach of direct laser sources. In modern XFEL facilities, self-seeding has become a practical route to generating more stable and spectrally pure pulse. Together, HGHG and self-seeding represent complementary approaches: the former excels where suitable external lasers exist, while the latter extends FEL coherence into the hard X-ray regime where external seeding becomes impractical.

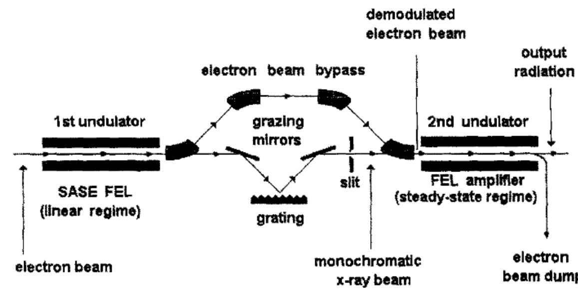

FIG. 8. Initial design proposal of the self-seeding process for the TESLA Test Facility at DESY by J. Feldhaus *et al.* in 1997 [20].

F. Bunch Compressors

XFELs can be used to emit ultra fast X-ray pulses. The duration of a pulse directly corresponds to the time it takes for the entire electron bunch to cross a certain point along the undulator. To acheive shorter pulses, bunch compressors are used to reduce the length of a bunch. Compressing an electron bunch is nontrivial, as the individual particles naturally repel each other via the Coulomb interaction. In practice, bunch compression is achieved by sending the electron bunch through a magnetic chicane. The chicane consists of a series of magnets that bend the electron bunch in a controlled manner, such that lower energy particles are deflected at greater angles, and thus must cover more distance, while higher energy particles are deflected less. A bunch is said to have a negative energy chirp if tail electrons have higher energy than head electrons. With a negative energy chirp, the tail electrons deflect less than the head electrons, allowing them to catch up to the head electrons, compressing the bunch [21]. Such energy chirps are often induced via off-crest acceleration in RF cavities. This is achieved by adjusting the phase of the RF field in the cavity such that the leading electrons enter the cavity when the field is at a trough of its standing wave, while the tailing electrons enter when the field is at a crest. In this manner, the leading electrons are decelerated while the tailing electrons are accelerated. You may wonder why such a system is not sufficient to compress the bunch on its own. The reason is that in this already relativistic regime, the energy chirp is not sufficient to significantly alter the velocity of particles. This is why chicanes are employed.

This bunch compression enables the emitted beam to reach femtosecond to sub-femtosecond time scales, thus allowing researchers to study the dynamics of both surface and sub-surface phenomena at nanometer or even atomic scales, where time-resolved experiments employing optical lasers, such as optical reflectometry, interferometry, and spectroscopy, suffer from limited spatial resolution and a lack of bulk sensitivity [22].

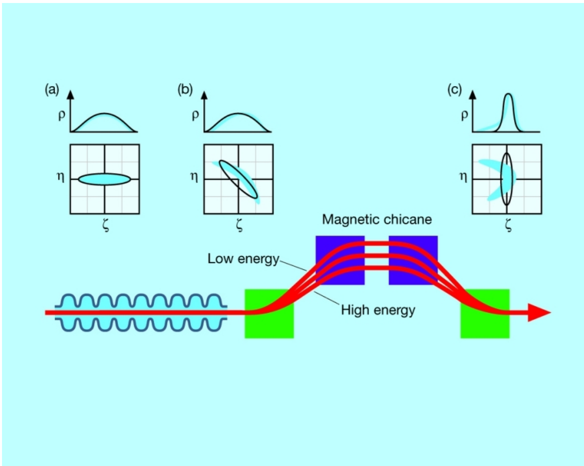

FIG. 9. Process of electron bunch compression via a magnetic chicane. A typical chicane will consist of four bending magnets which deflect particles of higher energy less then those of lower energy. In the plots (a), (b) and (c), we can see the longitudinal position (ξ) of the particles against their relative energy deviation (η), and charge densities (ρ). Initially, the bunch is spread out in space, with a uniform energy distribution (a) and wide charge density. After a negative energy chirp is induced, the tailing electrons are of higher energy than the leading electrons (b), but the density is unaffected. Finally, the bunch is compressed, with the tailing electrons catching up to the leading electrons (c) and the charge density is more peaked. [21].

G. Synchrotrons

Synchrotrons differ from XFELs in that they accelerate charged particles in a circular trajectory using bending magnets. The strength of the magnetic field must be synchronized with the particle's energy, as a greater Lorentz force is required to maintain the desired curved path as the particle's velocity increases. Due to this radial acceleration, the charged particles emit electromagnetic radiation. However, unlike in XFELs, this emitted radiation is largely incoherent. Synchrotron radiation has a broad spectrum, making it versatile in this sense. Some synchrotrons are called storage rings, as they maintain a beam of constant velocity particles circulating in a closed loop. In these facilities, radio-frequency cavities are used to compensate for the energy loss of the particles as they lose energy due to synchrotron radiation. Some synchrotrons have linear accelerators that provide an initial energy boost to the particles before they are injected into the storage ring. The emitted radiation is then directed to the experimental stations, where it is used for X-ray diffraction, spectroscopy, and imaging [23].

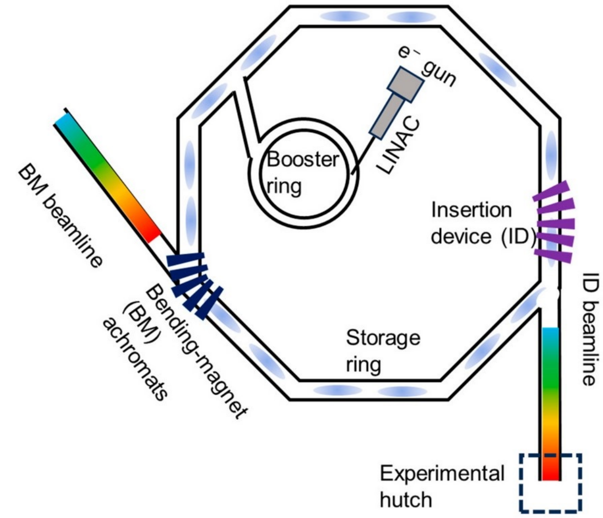

FIG. 10. Typical layout of a synchrotron facility. The linear accelerator (LINAC) is used to accelerate particles to high energies before they are injected into a booster ring. The booster ring then accelerates the particles to even higher energies before they are injected into the storage ring. The storage ring maintains a beam of particles circulating in a closed loop via bending magnets, where they emit synchrotron radiation. This radiation is then directed to the experimental hutch. Certain insertion devices, which may include undulators or wigglers, are used to generate higher intensity X-rays [23].

IV. DATA ACQUISITION AND ANALYSIS

Data acquisition in XFEL experiments is a complex process that involves detection of often ultra-fast X-ray pulses operating at high luminosities. The detectors used in these experiments must have high temporal resolution to capture the fast dynamics of the pulses, and a dynamic gain switching architecture to cope with said luminosities, while also being equipped with the ability to detect traditional laser light with wavelengths in the visible to infrared region with lower luminosities, allowing for pump-probe experiments (see Section ??). Said experiments require precise timing synchronization systems to ensure that the pump and probe signals are properly correlated. This typically requires a chain of mode-locked oscillators, pulse-stretchers and compressors, regenerative amplifiers and multi-pass amplifiers [24]. For detection, the European XFEL, for example, is outfitted with a Large Pixel Detector (LPD), constructed with 16 identical supermodules, each outfitted with a Front-End-Module which provides readout and control functions. Such a detector is expected to generate 10 Gb/s of processed data for every one Megapixel of sensor area [25]. It additionally uses an Adaptive Gain Integrating Pixel Detector, which is a hybrid detector with a 4.5 MHz frame rate, and a dynamic range capable of single photon resolution all the way up to a thousand 12.4 keV (X-ray energy) photons [26]. Other common types of detectors used are scintillation detectors, which convert X-rays into visible light, and semiconductor detectors, which convert X-rays into electrical signals, and charge coupled devices (CCDs). The latter are more sensitive and have a higher dynamic range, making them ideal for high-resolution studies. The data collected by the detectors is then processed using computational methods.

A. The Phase Problem

In the study of crystals by X-ray diffraction, the first step toward unveiling the atomic structure lies in the careful measurement of diffracted intensities. The intensity $I(hkl)$ of each diffraction spot is directly proportional to the square of the modulus of the complex-valued structure factor:

$$I(hkl) \propto |F(hkl)|^2 \tag{20}$$

The structure factor $F(hkl)$ encodes how all atoms within a unit cell contribute to scattering in a given direction. It is defined as:

$$F(hkl) = \sum_j f_j e^{2\pi i(hx_j + ky_j + lz_j)} \tag{21}$$

where the sum runs over all atoms j in the unit cell. Here, f_j is the *atomic scattering factor* of atom j, a measure of how strongly the atom scatters incident X-rays. It depends upon the number of electrons in the atom and the scattering angle, as more electrons yield greater scattering power. The coordinates (x_j, y_j, z_j) are the *fractional atomic coordinates*, specifying the position of atom j within the unit cell as a fraction of the unit cell lengths a, b, and c along the crystallographic axes. These positions, taken together, define the geometric and chemical structure of the unit cell and hence determine the resulting interference pattern observed in reciprocal space. This summation defines the complex structure factor, whose magnitude determines the measured intensity, while its phase, essential for structure reconstruction, remains experimentally inaccessible. Though X-ray diffraction patterns yield rich structural insight, it is cursed with a fundamental shortcoming known as the *Phase Problem*. Detectors record only the intensities of the diffracted beams, proportional to the square of the amplitude of the scattered waves, yet the phase of each wave is tragically lost in measurement. To fully determine a crystal's internal structure, one must reconstruct the electron density function, which denotes the real-space positions of electrons, and hence, atoms. The electron density function is given by

$$\rho(x,y,z) = \frac{1}{V}\sum_{hkl} |F_{hkl}| \cdot e^{-2\pi i[h\frac{x}{a} + k\frac{y}{b} + l\frac{z}{c} - \phi_{hkl}]}, \tag{22}$$

which is the inverse Fourier transform of the reciprocal-space structure factors. As we can see, phase information is essential for reconstructing the real-space electron density, and hence the Phase Problem is a central challenge in crystallography.

B. The Patterson Method

The challenges associated with measuring the relative phases among the diffracted beams ϕ_{hkl} makes determining the atomic positions within the unit cell difficult. Thankfully, in 1934, A. L. Patterson proposed a method to solve this problem. He showed that the one could replace the structure factors and phases in the electron density with the squared amplitudes of the structure factors alone [27]. This is beneficial, as the squared amplitudes can be directly measured from the diffraction pattern, and thus the phase problem in crystallography is avoided. The Patterson function is defined as:

$$P(u,v,w) = \frac{1}{V}\sum_{hkl} |F_{hkl}|^2 e^{2\pi i[hu/a + kv/b + lw/c]}. \tag{23}$$

One may recognize this as the Fourier transform of the electron density, save for the fact that the structure factors are squared. This is by consequence of the Wiener-Khinchin theorem, which states that the Fourier transform of an auto-correlation function yields the squared magnitude of the Fourier components of the original function. In this case, the Fourier transform of the Patterson function gives the squared structure factors, $|F_{hkl}|^2$, which are directly measurable from diffraction experiments. In this instance, the Patterson function is the Fourier transform of the electron density's auto-correlation function

$$P(u,v,w) = \int \rho(x,y,z)\rho(x+u, y+v, z+w)d^3\mathbf{r} \tag{24}$$

which describes how well $\rho(\mathbf{r})$ correlates with a version of itself shifted by an amount (u,v,w), in another words, a convolution with its mirror image. This symmetry is tied to intrinsic symmetry exhibited by crystals, and it can be shown that if one finds a maximum of $P(u,v,w)$, then at some point (u_1, v_1, w_1), then there are atoms (correspoding to maxima of $\rho(\mathbf{r})$) whose distance apart is given by the vector whose components are (u_1, v_1, w_1) [27]. The Patterson function displays a large number of peaks, specifically, for a crystal formed by N atoms in the unit cell, $P(u,v,w)$ will have N^2 maxima, corresponding to all possible interatomic vectors, including self-vectors (see Fig. 11). One can use the peak positions to determine the (x,y,z) coordinates of the atoms in the unit cell using symmetry operations, as was first demonstrated by David Harker in 1935 [28]. These peaks, qualitatively, have a wide profile, as the atoms in the unit cell are not exactly point-like. This may result in overlap among peaks. The heights of the peaks are proportional to the atomic number of the constituents of the unit cell, and thus heavier atoms will dominate the Patterson map, making for clear landmarks in the image. This method is complimentary to Rietveld refinement, as it can help establish the theoretical model for which the refinement process fits the data to. It is also generally quicker and simpler to perform, but yields less insight as a result.

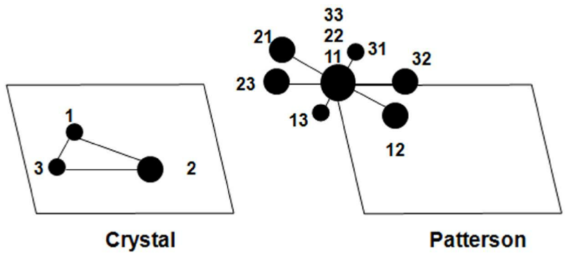

FIG. 11. Typical Patterson map of a crystal with three atoms in the unit cell. The peaks are labelled with their corresponding interatomic vectors

C. Rietveld Refinement

Rietveld refinement is a computational method used to determine the crystal structure of a material from XRD data. It is typically used for powder samples, where multiple reflections take place. It involves comparing the experimental diffraction pattern I_i^{obs} with a theoretical one I_i^{calc} generated from a proposed crystal structure. A general recipe for obtaining a theoretical model of a diffraction pattern is to model the diffraction peak positions, intensities, and shapes, taking into account background contributions.

Peak positions, as described in section II, are determined according to Bragg's law. For example, the orthorhombic crystal system has angles which obey

$$2\theta_{hkl} = 2\arcsin\left(\frac{\lambda}{2d_{hkl}}\right), \quad \text{where } d_{hkl} = \left(\frac{h^2}{a^2}+\frac{k^2}{b^2}+\frac{l^2}{c^2}\right)^{-1/2}. \tag{25}$$

Since the Miller indices h, k, l are integers, the peak positions are discontinuous, which properly models the observed peak behaviour.

The intensity of the diffracted X-rays is a function of the structure factor which depends upon the atomic positions, the multiplicity of the k reflection m_k, the temperature factor B_n, and the atomic form factor f_n. The form factor f_n is a measure of the scattering power of an atom in the sample, and is dependent on the electron density, the scattering angle and the scatterer. That is, X-ray form factors are different from neutron form factors, or electron form factors. For various techniques, these factors are often tabulated in the *international tables for crystallography* [29], alongside the multiplicity factors m_k, which refer to the number of equivalent reflections that arise due to a certain crystal class' symmetries. The temperature factor, also called the Debye-Waller factor, is a measure of the thermal vibrations of the atoms in the crystal. It accounts for the attenuation of scattered X-rays due to the thermal motion of the atoms [5]. Other factors which affects peak intensities include the Lorentz polarization factor L_k, which accounts for the polarization of the incident beam and preferred-orientation factors P_{kj}, which account for anisotropy in the orientation distribution of diffraction data, which is a formalism used to describe the texture of a material [30]. A polar diagram can be seen in figure 12, which plots diffraction intensity as a function of azimuthal angle and Bragg angle, clearly showing the importance of considering anisotropic diffraction properties when modelling.

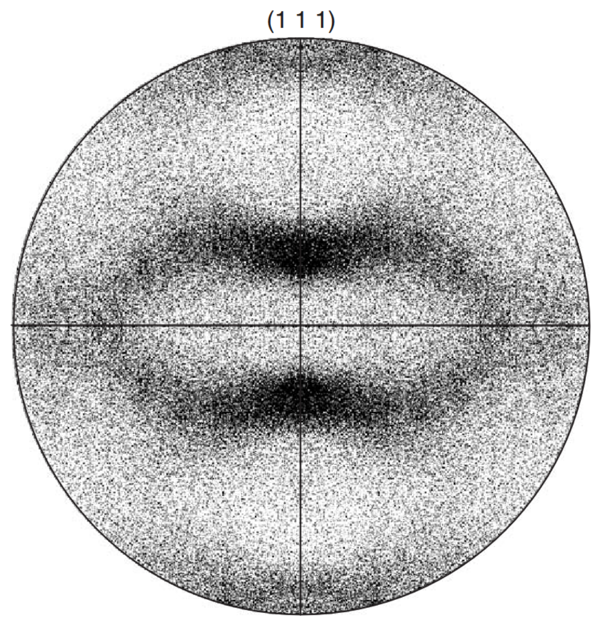

FIG. 12. Polar diagram of a rolled aluminium sheet for beverage cans [31]. Polar diagrams are graphical representations of how X-ray diffraction intensities vary with crystal orientation. They are used to visualize preferred orientation, texture effects and anisotropic diffraction properties.

The peak shape functions S_j are often modelled as a convolution of functions that account for broadening effects due to the instrumentation, wavelength dispersion, as described in the treatment of Eq III C, and the intrinsic properties of the sample. Common peak profiles include the Gaussian, Lorentzian, and pseudo-Voigt profiles. Absorption factors A_j, which account for the attenuation of the beam by the sample play a role in peak shapes as well. Finally, the background is often modelled

TABLE II. Factors affecting different components of an X-ray diffraction pattern.

Pattern component	Crystal structure	Specimen property	Instrumental parameter
Peak position	**Unit cell parameters** ($a, b, c, \alpha, \beta, \gamma$)	*Absorption* Porosity	**Radiation (wavelength)** *Instrument/sample alignment* Axial divergence of the beam
Peak intensity	**Atomic parameters** (x, y, z, B, etc.)	*Preferred orientation* Absorption *Porosity*	Geometry and configuration **Radiation (Lorentz, polarization)**
Peak shape	*Crystallinity* Disorder Defects	*Grain size* *Strain* *Stress*	**Radiation (spectral purity)** **Geometry** **Beam conditioning**

Key parameters are shown in **bold**. Parameters that may have a significant influence are shown in *italics*.

as a polynomial function in 2θ of degree N_b of the form

$$\mathrm{bkg}(2\theta_i) = \sum_{j=0}^{N_b} a_j (2\theta_i)^j. \tag{26}$$

Putting all these together, the model for the diffraction pattern is given by:

$$I_i^{calc} = S_F \sum_{j=1}^{N_p} \frac{\phi_j}{V_j^2} \sum_{k=1}^{N} L_k |F_{kj}|^2 \\ \times S_j(2\theta_i - 2\theta_{kj}) P_{kj} A_j + \mathrm{bkg}(2\theta_i) \tag{27}$$

where S_F is the beam intensity, ϕ_j is the phase volume fraction, and V_j is the phase cell volume, which is relevant for multiphase samples such as polycrystalline materials. L_k is the Lorentz factor, F_{kj} is the structure factor, S_j is the peak shape function, P_{kj} is the polarization factor, and A_j is the absorption factor. Factors which should be considered when modeling diffraction peaks can be found in II, which is taken from Pecharsky and Zavalij [2].

With the model in hand, the Rietveld refinement process involves adjusting the parameters to minimize the weighted sum of squares via

$$WSS = \sum_{i=1}^{N} \left[w_i (I_i^{\mathrm{obs}} - I_i^{\mathrm{calc}}) \right]^2, \\ \text{where } w_i = \frac{1}{\sqrt{I_i^{obs}}}. \tag{28}$$

Here, N is the number of observed data points. The process is iterative, and the parameters are adjusted until the theoretical pattern matches the experimental one. R indices are used to quantify the goodness of fit, defined as

$$R_{\mathrm{wp}} = \sqrt{\frac{\sum_{i=1}^{N} [w_i (I_i^{obs} - I_i^{calc})]^2}{\sum_{i=1}^{N} (w_i I_i^{obs})^2}} \tag{29}$$

$$R_{\mathrm{exp}} = \sqrt{\frac{(N-P)}{\sum_{i=1}^{N} (w_i I_i^{obs})^2}} \tag{30}$$

where P is the number of refined parameters. The goodness of fit is taken to be $R_{\mathrm{wp}}/R_{\mathrm{exp}}$, with values less than 2 indicating a good fit [32].

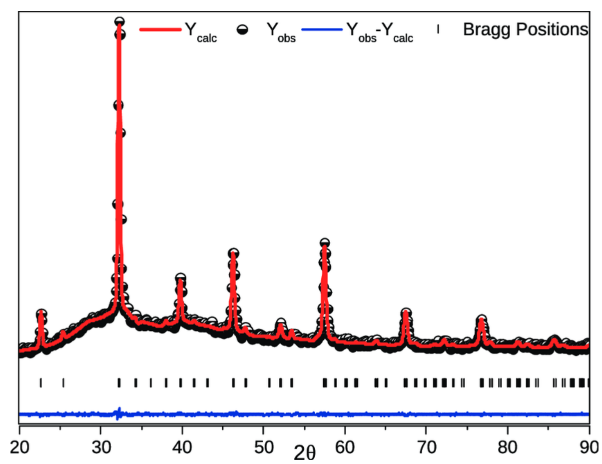

FIG. 13. Rietveld refinement plot of $LaMnO_3$ by P. Touseef and S. Shaibal [33]. The black circles are the observed data and the red line is the calculated spectrum via Rietveld refinement. The veritcal bars are the Bragg peak positions and the blue line shows the residuals.

Rietveld refinement has many uses, from determining sample purity via measureing phase fractions with high accuracy, to determining atomic positions, occupancies, thermal parameters and strain effects. It works with both X-ray and neutron diffraction data allowing for complimentary structural analyses. Unfortunately, these advantages come at the high cost of requiring a detailed model with multiple parameters. This complexity may lead to overfitting to the model, and the refinement process is often computationally expensive. Furthermore, this method is hightly sensitive, requiring good data quality, and only works for crystalline samples which have predicatable Bragg peaks, lest the model be drastically revised.

Common software packages for this technique include FullProf, GSAS and Reitan. These offer support for crystallographic information files (CIF), which are standard files used to package information about crystal models, such as atomic positions, unit cell parameters, thermal parameters and space group symmetry. Such files for common materials are often available from online databases [34].

V. EXPERIMENTAL TECHNIQUES

A. Serial Femtosecond Crystallography (SFX)

Many challenges arise when trying to measure diffraction peaks for nanocrystals. The main hurdle is radiation damage, as X-rays are simply too damaging and alter the structure of the specimen under study, yielding poor results. Cryogenic cooling of samples is often employed to reduce the rate of reactions within a sample, enabling it to withstand a 30-50 times increase in exposure [35]. This solution, however, complicates the sample preparation process, and can be undesirable if one wishes to study biological samples, for example, whose characteristics are wildly altered when cooled to this degree. XFELs offer an alternative. XFELs are capable of firing femtosecond flashes of X-rays that are shorter in time than it takes for atoms in the sample to move [36]. This is colloquially referred to as "diffraction before destruction". Do not let this phrase fool you; this technique will obliterate the specimen under study, as the intensity of such pulses are akin to concentrating the intensity of the Sun's rays upon the planet Earth directly onto this poor nanocrystal [37]. The power of this technique comes from the fact that the pulses themselves are so short in duration that the X-rays are diffracted faster than the radiation damage has time to manifest.

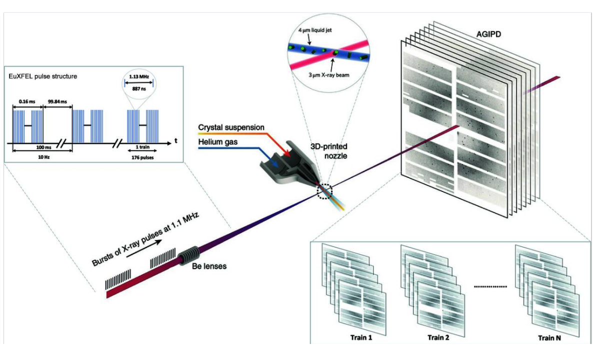

FIG. 14. Typical SFX experimental setup. A nozzle fires a liquid jet containing nanocrystals into the interaction point and enormous amounts of XRD images are generated. The EuXFEL pulse structure is shown. Boasting an effective repetition rate of 1.1 MHz, enough pulses are generated to meet the crystal orientation sampling requirements [38].

No longer is growing a large crystal out of a specimen a requirement for studying its diffraction patterns. Instead, the specimen nanocrystal is studied and destroyed in series with each X-ray sampling a different random orientation, and with enough sampling, impressive information about its structure is gleaned. This is often done suspended in solution, with the crystals floating into the line of fire.

An interesting case study for this method happened during the COVID-19 pandemic. The Linac Coherent Light Source (LCLS) awarded rapid access beamtime for seven SARS-CoV-2-related experiments in 2020 and 2021. These experiments aimed to obtain radiation damage-free protein structures at room temperature under physiological conditions, a key ability afforded by SFX. One notable study focused on the SARS-CoV-2 main protease (MPro), a critical target for drug development. The research team, based at KOÇ University in Istanbul, was the first to take advantage of this special beamtime call. They successfully solved the structure of MPro using SFX at the macromolecular femtosecond crystallography (MFX) instrument at LCLS. The team used the high-speed data collection capabilities of the facility to test nine different drug candidates. These experiments were performed using micrometer-sized crystals, which required the precision of XFELs to study without causing radiation damage. This study contributed to the ongoing effort to identify potential inhibitors for COVID-19 treatment and resulted in a peer-reviewed publication [39].

B. Single Particle Imaging (SPI)

Single-Particle Imaging (SPI) is a transformative method in structural biology, offering a solution for studying specimens that do not form well-ordered crystals, such as proteins, viruses, or macromolecular complexes. Unlike Serial Femtosecond Crystallography (SFX), which benefits from the regularity of crystalline lattices and Fourier synthesis, SPI faces the formidable challenge of retrieving phase information, which is lost in the absence of a crystal lattice. The phase problem is a significant obstacle, as non-crystalline samples produce noisy diffraction patterns that are difficult to interpret. In this method, the scattering signals are much weaker compared to the bright Bragg peaks seen in crystalline samples, and the data collection process is more demanding. Isolated particles are injected into the XFEL beam as a gas-phase stream, either through aerosol injection or electrospraying, under high vacuum conditions (typically 10^{-5} to 10^{-6} mbar). Each femtosecond pulse scatters off a single particle before radiation damage occurs, capturing a diffraction pattern that can be used for reconstruction. By computationally aligning thousands or even millions of such diffraction snapshots, researchers can generate high-resolution 3D models of the particles, bypassing the need for crystallization. This approach is particularly revolutionary for studying biomolecules that are otherwise challenging to crystallize, opening up new

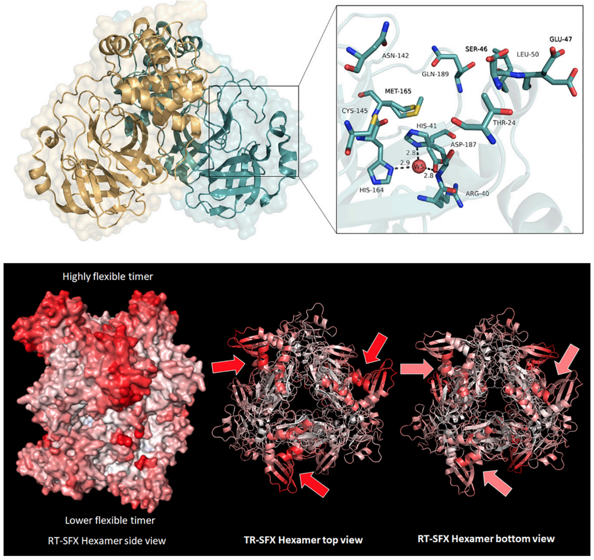

FIG. 15. SFX results for two key SARS-CoV-2 proteins: M^{pro} (main protease) and NendoU (endoribonuclease). M^{pro} is essential for processing the viral polyproteins into functional units required for replication and transcription. The structure obtained via SFX reveals detailed atomic interactions within the active site, including potential oxidative modifications, which may inform antiviral drug design. NendoU, or non-structural protein 15, plays a role in evading host immune responses by cleaving viral RNA intermediates. The high-resolution SFX structure highlights its catalytic core and coordination of metal ions, shedding light on its mechanism and vulnerability to inhibitors. These results were obtained using ultrafast XFEL pulses, allowing structure determination before radiation damage occurs, thereby capturing native conformations critical for drug development [39].

avenues for structural insights into a wide range of biological systems.

C. Machine Learning and Data Management

These techniques generate enormous amounts of data, as hundreds of thousands of diffraction peaks can be required to properly reconstruct the crystal structure, and among these images reside just as many images of absolutely nothing, as most X-rays simply miss the target altogether. Advancements in computer vision are proving to increase the efficiency of such methods. One way such advancements can benefit this technique is by data reduction through filtering out undesireable events, such as missed shots, from the pool of results immediately. For example, Rahmani et al. outline a method of data reduction for SFX experiments by employing the *Features from Accelerated Segment Test* (FAST), an efficient corner-detection algorithm for computer vision, allowing for the identification of Bragg peaks, combined with *Binary Robust Independent Elementary Features* (BRIEF), which is a state-of-the art description algorithm which takes as input the detected corners from FAST and generates compact binary strings from these keypoints using relative pixel intensity differences, thus encoding key features. Taking these two methods together, and making the corner-detection rotationally invariant, and scale invariant, we arrive at the *Oriented FAST and Rotated BREIF* (ORB) method.

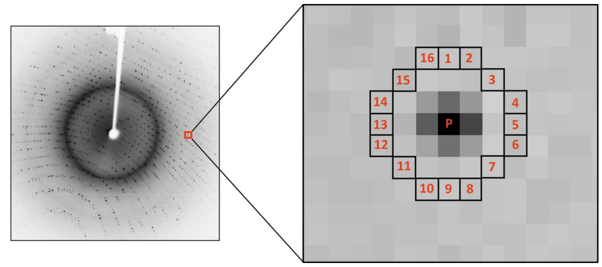

FIG. 16. The FAST key point detection mechanism. A pixel p is compared with 16 neighbouring pixels in a circle. If more than eight pixels are darker or brighter than p then it is identified as a Bragg peak [38].

Machine learning architecture advances have similarly led to the use of multilayer perceptrons to classify the images using the ORB descriptions according to more nuanced categories such as weak hits, strong hits, background dominated images, etc. Further, advancements in convolutional neural network architecture has led to machine-learning driven crystal structure identification [40].

D. Pump-Probe Experiments

The singular temporal resolution afforded by XFELs has opened the gateway to pump-probe experiments, wherein the temporal dance of matter is laid bare with femtosecond precision. In this method, a sample is first excited by a well-timed optical pulse, called the pump, often in the visible or infrared region, perturbing the system from its equilibrium. Then, after a precisely chosen delay, the sample is struck by a pulse of X-rays - *the probe* - which captures a structural snapshot at that instant.

By repeating this process and altering the time delay between the pump and the probe, a given specimen's response to the laser pulse can be measured at any point along the process. With this, one can image molecules undergoing chemical reactions, phase transitions, structural reformations, enabling the creation of ”real-time” movies. Of course, this too destroys the sample after each ”frame”, so scientists are cycling through numerous ”protagonists” to shoot these films. This is a far cry from traditional methods often limited to such studies in thermal or chemical equilibrium. An example of this technique is Yungok Ihm *et al.* imaging gold nanoparticles melting. In this experiment, the pump laser served to heat gold nanoparticles and the probe measured the phase transition dynamics of solids melting.

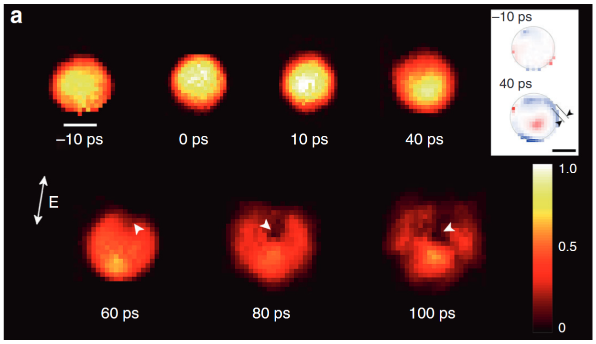

FIG. 17. Time-resolved single-shot images of single gold nanoparticles undergoing melting at various time delays, ranging from 10 picoseconds before the probe, to 100 picoseconds after the onset of melting. The arrows show the polarization direction of the pumping laser, which is shown in this paper to affect where the starts and how it flows in otherwise symmetric gold nanoparticles [41].

This experiment, among others, sheds light on the fundamental mechanisms behind solid melting which have remained elusive at the nanoscale. Another prominent example utilizing this technique is the study of Photosystem I (PSI) by Chris Gisriel et al. a key protein-pigment complex involved in the light reactions of photosynthesis. Located in the thylakoid membranes of plants, algae, and cyanobacteria, PSI captures solar energy and drives electron transfer from plastocyanin to ferredoxin. This electron transfer is crucial for producing NADPH, which in turn powers the Calvin cycle and the synthesis of carbohydrates [42]. Using femtosecond pump-probe techniques at XFEL facilities such as the European XFEL, researchers have successfully observed the ultrafast structural changes within PSI during photoactivation. By triggering PSI crystals with a visible light pulse and probing them with tightly synchronized X-ray pulses, they captured transient intermediates in the electron transfer chain with picosecond resolution. These snapshots illuminate how PSI initiates charge separation and rapidly funnels electrons through its cofactors, providing a detailed picture of its exceptional quantum efficiency. Understanding these ultrafast processes is not only of fundamental biological interest but also has broader implications. Photosynthesis represests one of nature's most efficient systems for solar energy conversion. By uncovering its underlying mechanisms, scientists hope to inform the design of artificial photosynthetic systems and next-generation solar energy technologies.

VI. NEXT STEPS

While XFEL-based experiments have enabled transformative advances in structural biology and materials science, several challenges temper their widespread adoption. Chief among these are the limited beamtime availability and the high operational and maintenance costs

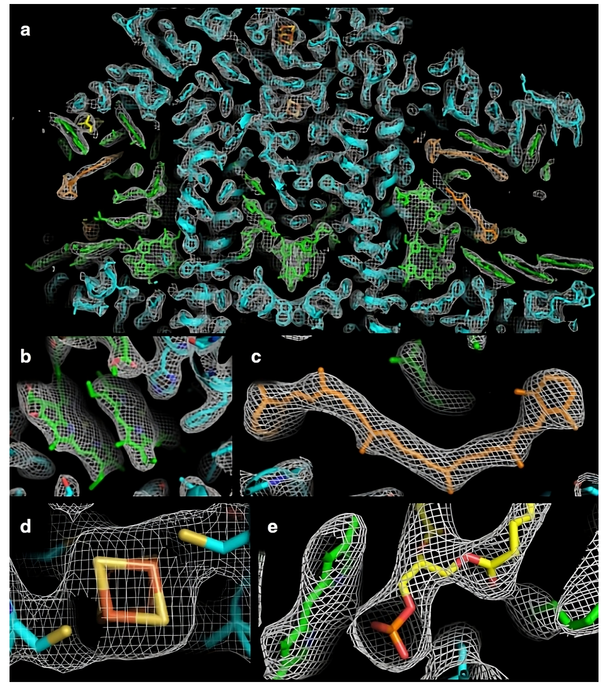

FIG. 18. Electron density map (2Fo-Fc at 1.5σ) and model of structural elements in the XFEL-resolved Photosystem I (PSI) monomer. (a) Slice through the electron density of the full PSI monomer. (b) Close-up of the "special pair" of chlorophylls (P700) involved in primary light absorption. (c) A β-carotene molecule, important for photoprotection and structural stability. (d) The 4Fe-4S cluster FX, which transfers electrons within PSI. (e) A phosphatidylglycerol lipid interacting with a chlorophyll molecule via axial coordination.

associated with large-scale XFEL facilities. Access to beamtime is highly competitive, often necessitating long lead times and meticulous proposal processes. Moreover, the sheer complexity of these facilities—requiring superconducting accelerators, kilometer-scale undulator arrays, and cryogenic systems—renders them expensive to construct, operate, and maintain. To address these limitations, the next frontier in XFEL development lies in the pursuit of compact XFEL sources. One promising direction involves the use of shorter undulator periods in combination with high-brightness, low-emittance electron beams. By reducing the undulator period, the total undulator length required to reach X-ray wavelengths can be significantly shortened, thereby decreasing the physical footprint and cost of the facility [43]. Technological advancements in laser-driven photocathodes, high-gradient accelerators, and beam transport optics are crucial to realizing such systems. Another rapidly emerging approach is inverse Compton scattering (ICS), wherein relativistic electrons scatter off optical laser photons to produce high-energy X-rays. Unlike conventional XFELs, ICS-based sources do not require undulators at all, and their compact nature makes them attractive for laboratory-scale implementations [44]. While current ICS systems lack the brightness and coherence of traditional XFELs, ongoing developments in laser and beam synchronization, pulse shaping, and electron beam quality offer promising paths toward closing this gap. In summary, while large-scale XFELs remain the pinnacle of ultrafast, high-resolution structural imaging, the development of compact, cost-effective XFEL alternatives could democratize access and catalyze a broader spectrum of applications, from hospital-based medical imaging to in situ studies of catalytic processes and beyond.

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